

Exploring Cloud Computing Stack — Simple & Structured Explanation

Cloud computing is often described as a “stack” because it is made up of layers of services built on top of one another. These layers help users and companies run applications, store data, and access resources through the internet.

1. What is Cloud Computing? (NIST Definition — Simple Explanation)

According to NIST (National Institute of Standards and Technology), cloud computing is:

👉 *A model that gives on-demand access to a shared pool of resources, such as:*

- *servers*
- *networks*
- *storage*
- *applications*
- *services*

These can be accessed quickly, with minimal management effort by the user.

In simple words: Cloud = Fast access to IT resources over the internet without managing the hardware yourself.

2. Why is it called a “Stack”?

Because cloud computing has different service layers, each built on top of the other:

- **IaaS** → Bottom layer
- **PaaS** → Middle layer
- **SaaS** → Top layer

Each layer provides different kinds of cloud services.

3. Three Main Types of Cloud Services (Cloud Service Models)

A. SaaS – Software as a Service

- Top layer of the cloud stack
- Provides ready-to-use applications for end users
- Accessed via the Internet
- No need to install or maintain anything
- **Examples:** Gmail, Google Docs, Zoom, Salesforce
- **Who uses it?** General users and businesses.

B. PaaS – Platform as a Service

- Middle layer
- Provides a platform with tools for developers
- Used to build, test, and deploy applications efficiently
- Developers don't manage underlying hardware
- Helps in coding and implementation
- **Examples:** Google App Engine, Heroku, Microsoft Azure App Services
- **Who uses it?** Software developers.

C. IaaS – Infrastructure as a Service

- Base layer / Foundation of the cloud stack
- Provides virtual hardware and underlying resources like:
 - servers
 - storage
 - networks
 - operating systems
- Users manage the OS and applications, cloud provides the infrastructure

- **Examples:** AWS EC2, Google Compute Engine, Microsoft Azure VMs
- **Who uses it?** System administrators, DevOps teams, and tech companies.

4. Essential Features of Cloud Computing (NIST Features)

These features must be present for a service to be considered “cloud computing”:

1. ★ **On-Demand Self-Service**

Users can sign up and use resources instantly, without waiting.

2. ★ **Broad Network Access**

Accessible from multiple devices: laptops, desktops, mobiles.

3. ★ **Resource Pooling**

Many users share the same physical resources, but they are isolated virtually.

4. ★ **Rapid Elasticity**

Resources can expand or shrink automatically based on demand.

5. ★ **Measured Service (Pay-as-you-use)**

Billing works like electricity/water — you pay only for what you use.

5. Everyday Usage of Cloud Services

Almost everyone uses cloud services indirectly or directly.

Examples:

- Using GitHub to manage code
- Using Google Drive or OneDrive
- Using Instagram / YouTube / Netflix (they run on cloud)
- Companies using cloud to run apps, databases, and websites

Businesses are moving all operations towards service-oriented cloud models.

6. Why Do People Use Cloud?

- Quick access to resources
- No hardware management
- Cost-efficient (pay only for usage)

- Scalable
 - Secure and reliable
-

1. Exploring SaaS (Software as a Service)

What is SaaS?

- SaaS = Software delivered over the internet.
- You access it through a web browser.
- The vendor (provider) manages everything: installation, updates, security, storage.

How is it licensed?

- Usually based on subscriptions or pay-as-you-use.
- Sometimes free, earning revenue through advertisements.

Key Features

- Managed centrally (users don't install anything).
- One-to-many model → one software serves many users.
- Users don't worry about patches, updates, or maintenance.

When should SaaS be used?

Use SaaS for applications like:

- **“Vanilla” applications**
Basic tools used by all companies (e.g., email). Does not give a competitive advantage.
- **Apps that depend on internet access**
Example: sales management software.
- **Apps involving external interaction**
Example: email newsletter campaign software.
- **Apps with sudden demand spikes**
Example: tax filing software, billing software.
- **Short-term applications**
Example: collaboration tools for a temporary project.

When NOT to use SaaS

Avoid SaaS for:

- Applications needing real-time, fast processing.
- Applications where laws don't allow storing data outside the company.
- Applications already having a strong on-site solution meeting all needs.

2. Exploring PaaS (Platform as a Service)

What is PaaS?

A platform that helps developers build, test, deploy, and manage applications without buying hardware or maintaining infrastructure.

How PaaS differs from SaaS

- **SaaS** = ready-made software delivered to end users.
- **PaaS** = platform/tools for developers to create new software.

What PaaS includes

PaaS typically offers:

- Tools for developing, testing, deploying, and maintaining apps.
- Web-based UI creation tools.
- Billing management tools.
- Built-in scalability, failover, and load balancing.
- Integration with databases and web services.
- Multitenant architecture (many users using same application).
- Support for team collaboration (project planning tools).

When should PaaS be used?

- When many developers collaborate on the same project.
- When external parties must interact with the development process.

- When organizations have existing data sources and want to build applications on top of that data.

When NOT to use PaaS

Avoid PaaS when:

- Applications must be portable (PaaS may lock you to one provider).
- Application uses proprietary languages that can restrict development.
- Application needs custom hardware/software for high performance.

3. Exploring IaaS (Infrastructure as a Service)

What is IaaS?

IaaS gives virtualized computing infrastructure such as:

- servers
- storage
- networks
- operating systems

Users rent these resources on-demand instead of buying hardware.

Types of Cloud in IaaS

- **Public cloud:** Shared resources available through the internet.
- **Private cloud:** Dedicated infrastructure offered via a private network.
- **Hybrid cloud:** Combination of public + private cloud.

Key Features of IaaS

- Resources offered as services.
- Dynamic scaling available.
- Variable cost model → utility pricing (pay for what you use).

When should IaaS be used?

- When demands are changing frequently.
- For new businesses with low capital to buy hardware.
- When an organization is growing fast and scaling hardware is difficult.
- When companies want to shift from capital cost → operational cost.
- For short-term infrastructure needs.

When NOT to use IaaS

Avoid using IaaS:

- When regulatory compliance makes data outsourcing difficult.
- When on-site infrastructure can handle all organizational requirements comfortably.

Exploring IaaS, PaaS, SaaS – Simple, Structured, Complete Explanation

Cloud computing has three main service models:

- IaaS – Infrastructure as a Service
- PaaS – Platform as a Service
- SaaS – Software as a Service

They differ mainly in control, cost, and who manages what.

1. Key Difference: Control vs Cost

SaaS (Least control, lowest involvement)

- Vendor controls the application, OS, storage, networking.
- You only use the software.
- Best for small businesses or users who don't want to manage IT.

PaaS (More control than SaaS, less than IaaS)

- You control application + data.

- Vendor manages OS, runtime, middleware, infrastructure.
- Best for developers building apps without managing servers.

IaaS (Most control, highest responsibility)

- You control applications + OS + middleware + data.
- Vendor provides the hardware infrastructure.
- Best for companies needing full flexibility and scalability.

2. IaaS vs PaaS vs SaaS — Table Explanation

Attribute	IaaS	PaaS	SaaS
Overview	Provides virtual servers, storage, networking	Provides platform/tools to build and deploy applications	Provides ready-made software online
Responsibility	User manages apps, data, runtime, middleware, OS	Vendor manages runtime, middleware, OS; user manages applications	Vendor manages everything (apps, runtime, OS, infra)
Scalability	Highly scalable; user controls scalability	Auto-scaled by platform	Fully managed scalability by provider
Flexibility	Highest flexibility	Medium flexibility	Least flexibility
Development Time	Longest (you handle infrastructure)	Medium (focus on coding only)	Fastest (just use the software)
Example Providers	AWS, Azure, Google Cloud	Heroku, Google App Engine, Azure App Service	Salesforce, Google Workspace, Microsoft 365

3. Cloud Computing Components

Cloud computing uses several building blocks:

1. Compute

- Virtual machines
- Containers
- Serverless functions

2. Storage

- Object storage (e.g., S3)
- Block storage
- File storage

3. Networking

- Connects VMs, services, applications securely.

4. Databases

- Managed SQL/NoSQL databases.

5. Security

- Access control (IAM)
- Encryption
- Compliance tools

6. Management Tools

- Monitoring
- Logging
- Resource optimization

4. Cloud Computing Statistics (Important for understanding trends)

Market Trends

- Cloud market expected to reach \$1 trillion by 2026.
- Public cloud spending growing at 16.7% CAGR.

Adoption Rates

- 94% of enterprises use cloud services.
- Hybrid cloud adoption reached 58%.

Popular Cloud Providers

- Amazon AWS – 32%
- Microsoft Azure – 20%
- Google Cloud – 9%

Security Concerns

- 66% of IT teams say security is the top concern.
- Cloud-based DDoS attacks increased by 967%.

5. Extra Notes on IaaS (Important points)

- Gives complete control over infrastructure + applications.
 - Vendor spends on physical servers, networking, storage, not you.
 - Slightly costlier than PaaS/SaaS because you control more.
 - Useful when:
 - You're growing fast and need scalable hardware
 - You want to shift from capital cost → operational cost
 - You need temporary or project-based infrastructure
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1. Large Business Cloud Model (Hybrid Cloud)

Why Hybrid Cloud suits large-scale businesses

Large businesses prefer hybrid cloud because it combines:

- **Public cloud** → cheaper, scalable
- **Private cloud** → secure, controlled

This gives companies the best of both worlds.

How Hybrid Cloud works

Public cloud is used for:

- Non-sensitive data
- Email
- File sharing
- Web hosting

Private cloud is used for:

- Sensitive data
- Applications with strict security
- Systems requiring compliance (banks, healthcare)

Benefits of Hybrid Cloud

- **Cost efficiency** → Pay only for required public cloud resources.
- **Scalability & flexibility** → Increase or decrease resources based on demand.
- **Better security** → Private cloud protects critical data.
- **Agility** → Quickly respond to business changes.
- Access to advanced public cloud services like:
 - Data analytics
 - Machine learning
 - IoT
 - Serverless computing

2. Cloud Service Providers (CSPs)

What CSPs do

A Cloud Service Provider rents out:

- Technology
- Infrastructure
- Expertise

for cloud-based services such as:

- Online storage
- Computing power
- Networking

CSP Infrastructure

CSPs build and manage:

- Data centers
- Servers
- Hard drives
- Cooling systems
- Networking hardware

How users access CSPs

- Anyone can connect to cloud data centers via the internet
- Pay as you go (usage-based pricing)

3. Connecting to a Cloud Server

There are 4 main methods to connect to a cloud server:

A. OpenSSH (Linux/macOS/WSL)

What is SSH?

SSH = Secure Shell, used for safe login to remote Linux servers.

Command format

```
ssh username@remote-host
```

Example: `ssh root@your-public-ip`

Authentication

- If an SSH key is set → system uses key automatically
- If it's the first connection, you must type `yes` to confirm authenticity
- If no SSH key → system asks for OTP (one-time password)

After login

- You must change OTP to a new password
- Password characters don't appear on screen (security feature)

B. PuTTY (For Windows Users)

Windows doesn't have built-in SSH → PuTTY is used.

Steps to connect

1. Open PuTTY
2. Enter public IP in **Hostname**
3. Go to **Connection** → **Data** → Enter username (`root`)
4. Go to **Connection** → **SSH** → **Auth** → **Credentials**
5. Browse and select private key
6. Save session configuration
7. Click **Open** → Enter passphrase (if key has one)

If no SSH key:

- Enter server IP
- Click **Open**
- Use password login

C. Remote Desktop Connection (For Windows Servers)

Usage

For connecting to Windows Cloud Servers like a normal desktop.

Requirements

- Windows RDP client (built-in)
- On macOS/linux → use Remote Desktop app / Remmina

Login details

- IP Address
- Username: Administrator
- Password: given during deployment

Supports: Full-screen mode, Saving credentials (if system is secure)

D. Console Connection (HTML5 Web Console)

Used when:

- SSH fails
- Firewall issues
- System errors prevent normal login

How to use

1. Open Cloud Control Panel
2. Go to **Console** tab
3. Click **Open Console**
4. Login with `root` or other usernames

E. VNC Console

Another emergency access method through a VNC client.

Steps

1. Go to server settings → **Console**
2. Enable VNC service
3. Set VNC password

4. Save changes

```
}
```

Connect using: `hostname:port`

Example: `fihe12.vnc.upcloud.com:12345`

Note: Changing keyboard layout requires server shutdown